

1. INTRODUCTION

Data acquisition is collecting data or information that describes the given situation. Data collection is important for any industrial application. The collected data may be processed and delayed at the site itself so that decisions maybe made. Sometimes data acquisition may not be possible by traditional methods, as the site may not be easily accessible due to geographical reasons or the site may be hazardous to human life. Thus remote collection of data and control of the source is of prime importance in such a situation; this can be done either using wired or wireless mode of communication.

All industrial applications require collection of data in development, production, quality control management, process control, etc. The data collected is used to make decisions and control the operation by the computer without any human interface as per requirements i.e. automation helps to reduce human errors evolved in process applications. Because of automation decision is achieved resulting into perfection of received data. The complete system is divided into two parts-Hardware & Software. The hardware unit is further split into two units-Transmitter & Receiver & Controller.



Figure 1.1 Industrial Automation

Industrial Internet of Things (IOT) is the best way of connecting industrial machineries and sensors, to each other, over the internet, allowing the authorized user of the industry to use information from these connected devices to process the obtained data in a useful way. IOT-connected applications typically support data acquisition, aggregation, analysis, and visualization. The IOT architecture includes latest technologies such as computers, intelligent devices, wired and wireless communication and cloud computing. Previously Bluetooth and RF (Radio Frequency) technologies were used to control and monitor the industrial applications but were limited to short distance. The operator had to be in the range of the Bluetooth connectivity or in the Radio Frequency area. Solution to the short distance communication is the IOT based industry automation. Here we can have controlling as well as monitoring from anywhere in the world. The following Fig 1.1 illustrates IOT based Industrial Automation.

In any industrial processes there are several parameters to be measured and controlled. This measurement and controlling process is carried out from the main central control room where all the data from different locations are acquired. The data acquisition systems developed can use PC as a human machine interface using visual basic software and controlling can be done by relays. In recent years, PC I/O interface has become increasingly reliable, accurate and affordable. Because of this, PC based data acquisition control systems are widely used in industrial and laboratory applications. If control could be achieved at the click of a button it would lead to an enormous saving of manpower and a corresponding reduction in cost. This system can be used to monitor up to eight such parameters like Temperature, Humidity, Gas, Light Intensity etc.

1.1 What Is Data Acquisition System?

- Data acquisition (DAQ) is the process of measuring an electrical or physical phenomenon such as voltage, current, temperature, pressure, or sound with a computer.
- A DAQ system consists of sensors, DAQ measurement hardware, and a computer with programmable software.
- Compared to traditional measurement systems, PC-based DAQ systems exploit the processing power, productivity, display, and connectivity capabilities of industry-standard computers providing a more powerful, flexible, and cost-effective measurement solution.

1.2 Objectives:

Our project is developed around the Arduino UNO ATmega328. The Arduino was chosen because of its versatility and ease of availability. The main function of the microcontroller is to read the various parameters and send it via wireless communication. At the remote location the data is read by a personal computer. The equipments are also controlled by sending appropriate commands to the controlling unit using visual basic via parallel ports.

Internet of Things (IOT) is rapidly increasing technology. IOT is the network of physical objects or things embedded with electronics, software, sensors, and network connectivity, which enables these objects to collect and exchange data. Here, we are developing a system which will automatically monitor the industrial applications and generate Alerts or take intelligent decisions using concept of IOT. IOT has given us a promising way to build powerful industrial systems and applications by using wireless devices, Android, and sensors. A main contribution of this is that it summarizes uses of IOT in industries to monitor and control the Industry.

The main objective of the project is to acquire data from far off locations and from places not accessible to human beings and transmit this data through wireless communication. The entire working of the project is split into two parts viz. hardware and software data i.e. parameters such as temperature, pressure, etc. are converted into their corresponding electrical values and are applied at the input of Arduino UNO analog pins These analog values are converted into equivalent digital values by the inbuilt ADC.

1.3 Aim Of The Project

As it has been introduced before, the main objective of this project is to design and implement a system which can measure and process different parameters such as temperature or light by using the Arduino platform and SQL software, with the corresponding tests.

It is expected that the system allows the user to control in a very easy way the microcontroller by using a computer, with no need of having a deep knowledge about programming in Arduino. The user will be able to read the information and control the system by using SQL Software, which is intuitive and comfortable to use.

The way on how the system measures and controls the parameters is one of the main goals. The user of the system cannot receive any wrong data, or data with poor precision, because that would mean a failure on its application. Therefore, no mistakes are allowed as it can affect the process in which this system may be used.

Moreover, one more objective is that once the sensors are installed, it is easy to access and get the information the sensors provide for the user. This means that you will not need to be an expert on programming if you want to use it. SQL Software will help with this objective. In one word, intuitive.

Versatility in this project will also be one of the characteristic of this Arduino based data acquisition system. It will be possible to measure and process several parameters, not only a specific one and above everything, be useful for general purposes. Basically, the components that form part of the system are:

- Arduino Uno
- Different sensors
- Arduino software
- SQL Software
- VB.NET

1.4 History

First time automation in industries was done through the use of steam and water power. As the advancement took place, electricity was introduced and was used in industries for mass production. Fig 1.4.1 shows the use of steam power in industries in olden days. This machine was used in the first safe and successful steam power plant introduced by Thomas Newcomen in 1712.

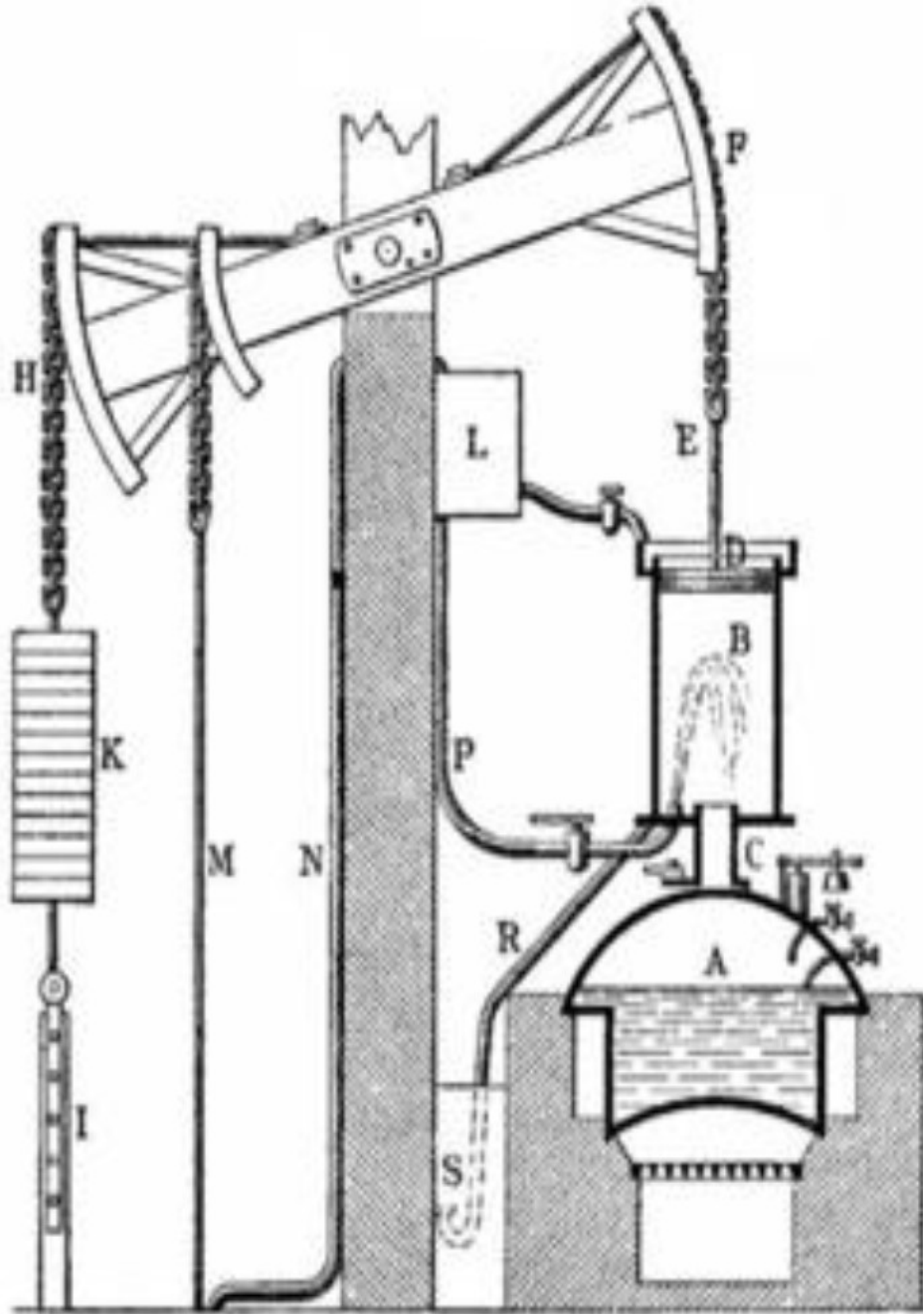


Figure 1.4.1 Newcomen's atmospheric steam engines

When computers were invented, it was designed to perform multiple functions. As time went on, computers became cheaper and then almost all industries started using it for automation because it reduced a major work load experienced by the humans and still it is considered as the best option to control and monitor a application. Automation is done through the

technologies such as Bluetooth and radio frequency which can be employed for short distance communication



Figure 1.4.2 RFID based automation in manufacturing industry

Automation through IOT can help to get rid of the short distance communication. Thus, introducing internet in industries can help to have control over the application from anywhere in the world. The following Fig 1.4.3 illustrates how IOT can help to ease the life in industries as well as in other fields.

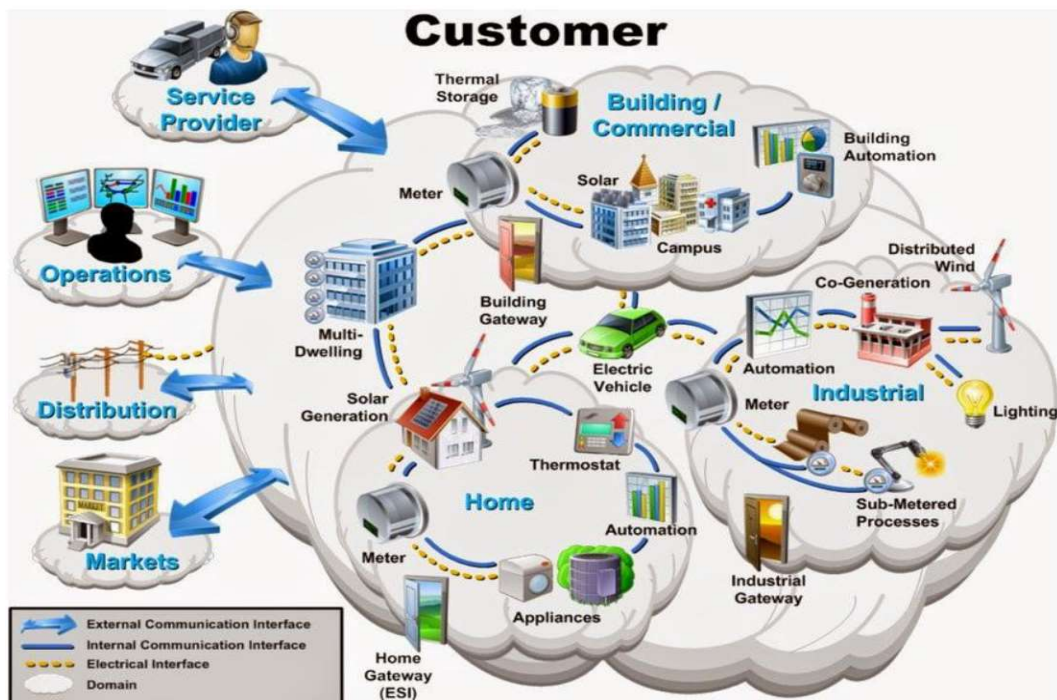


Figure 1.4.3 IOT in different fields

2. LITERATURE SURVEY

2.1 Literature Review:

- **Vehicle Tracking Using a Reliable Embedded Data Acquisition System With GPS and GSM.(Krishna Kishore, Sasi Vardhan, Lakshmi Narayana in 2010).**

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In this paper, they present the principles of a low operational-cost but flexible Internet-based data-acquisition system. The main core of the system is an embedded hardware running a scaled down version of Linux: a popular choice of operating system for embedded applications. The embedded device communicates through General Packet Radio Service (GPRS), which makes it accessible from anywhere in the world through a web server built into the embedded device. In addition, GPRS provides a bidirectional real-time data transfer allowing interaction. The proposed system eliminates the need for server software and maintenance.

GSM and GPRS are developed for cellular mobile communication. A GPRS connection with unlimited duration of connectivity is charged only for the data package transfers and adopted in several mobile remote control/access systems. GPRS becomes a cost-effective solution only if the data transfers can be optimized. Once a GPRS connection has been established, queried data can be relayed to the client via a central server. Using a central server to relay the acquired data has some disadvantages. First, a central server needs a client interface framework. An additional data transfer corresponds to time delays before the data are made available to the client. In addition, since the server acts as a relay, no direct bidirectional communication between the client and the embedded system can be established. This makes the system unsuitable for real-time control applications. The basic idea behind real-time processing is that the embedded system is expected to respond to the queries in time. Real time should be fast enough in the context in which the system is operating and reliable as well. Real-time system correctness depends not only on the correctness of the logical result of the computation but also on the result delivery time.

In this application, a low-cost, Internet-based data acquisition and control system has been designed and implemented that should find interest from researchers. The application possibilities are virtually unlimited by attaching modules with appropriate interfaces, although the usage of the system is demonstrated with only a few sample devices. Compared with other applications, this system has advantages in terms of allowing direct bidirectional communication and reducing overhead, which can be vitally important for some real-time applications. The operational costs have been reduced by relinquishing the storage of large data to an FTP server on the Internet. The system is designed to support both static and dynamic IPs. A method to distribute the IP information has been developed. This cost-minimization effort is a big concern for mobile systems using wireless communication methods and has not been discussed before. The overall cost advantage of the system in terms of the components used makes it an attractive choice for data-acquisition applications. The power demand of the device is still in the process of being improved by putting the attached devices into sleep mode at times when they are not in use to conserve power.

- **Design of DACS (Data Acquisition and Control System) using Linux (Stephen Nischay, Latha in 2012).**

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Associate Professor, Aurora Technological Research Institute, JNT University, Hyderabad

This designed can be used with industrial equipments medical instruments, and in many other places. A remote user can monitor and control the equipments with a simple but enhanced and more powerful user interface without additional hardware. Software contention and architectures can significantly affect web server performance. Poorly designed and configured software architectures might even generate high response times while the physical resources display low utilization. A remote user only requires a common internet browser to carry out experiments on real hardware.

Web servers bring the desired HTML pages over the world wide Internet or a local network to the web browser. With the development of industrial modernization, embedded technology has been widely applied in these areas. To meet the needs of these new conditions, designing a new monitor control system that can adapt to modern production becomes a top priority, and embedded system become the first choice. Based on S3C2440A ARM-9 core processor and transplanted Linux operating system, between database stations and server using C/S communication mode, between remote user and the server using B/S communications mode, in this paper we designed a kind of embedded system with integrated functions of data acquisition, data display, remote monitor and control.

The EWS that has been designed can be used with industrial equipments medical instruments, and in many other places. A remote user can monitor and control the equipments with a simple but enhanced and more powerful user interface without additional hardware. Software contention and architectures can significantly affect web server performance. Poorly designed and configured software architectures might even generate high response times while the physical resources display low utilization. A remote user only requires a common internet browser to carry out experiments on real hardware.

Rapid development in the field of industrial process control demands for high data accuracy and reliability. This Embedded ARM System can adapt to the requirements as a whole. By making use of SQ LITE - An embedded database, the collected data can be stored on the server and can be used for further processing. As ARM core is a fully accustomed and provided with USB Host device video processing can also be implemented.

- **Smart Real Time Embedded Arduino Based Data Acquisition System (Saraswat Teli, Mani in 2015).**

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In this paper, an effort has been made to design and develop a smart real time embedded Arduino based data logger for indoor and outdoor environment. Over here the designed product is much cost effective compared to existing products. The system is fully automated and once it is setup no human intervention is required for monitoring. This will also reduce errors in measurements as because many times human intervention leads to errors. Designed system consumes very less power so this is also power efficient and can function autonomous with small dedicated battery for backup with grid connection even in power outages.

The work has been concentrated to environmental parameters such as temperature, humidity and solar isolation. The data logger is proposed to be developed with the use of Arduino Uno based on ATmega328. The Arduino Microcontroller board is used which has inbuilt ADC and other peripheral circuitry necessary for operation. The physical parameter is sensed by the sensors and is converted into analog signal. This analog signal is fed to the Arduino board ADC pins which is then converted in to an equivalent digital quantity and is further processed in the microcontroller. The raw digital signal or processed signal out of microcontroller may be displayed on the LCD display or it's saved in a database or even at the same time this data is sent to computer through the USB serial port. The serial port data is then accessed and is imported in Microsoft excel for computation and graphical representation.

Designed product is much cost effective compared to existing products. Product is industry ready for implementation. The system is fully automated and once it is setup no human intervention is required for monitoring. This will also reduce errors in measurements as because many times human intervention leads to errors. This system can be altered marginally to log and record other physical signals rather than temperature, humidity or pressure in requisite application. Product can be custom build to suit prerequisite accuracy and precision of measurement by upgrading the analog to digital converter rather than using inbuilt 10 bit converter embedded in microcontroller. Data acquired can be imported further into computational software's like MATLAB, Python for mathematical computations. Multiple channel data acquisition can be implemented using collective number of sensors connected to analog or digital inputs respectively as per sensing parameter and sensor specifications.

2.2 Research Gap:

Krishna Kishore's paper - Internet based DAQ system allows direct bidirectional communication and reducing overhead, which can be important for some real-time applications. The operational costs have been reduced by relinquishing the storage of large data to a server on the Internet.

Saraswati Teli's paper - Arduino based DAQ system involves two steps operations like data monitoring as well as data logging. The system is fully automated and once it is setup no human intervention is required for monitoring. This will also reduce errors in measurements as because many times human intervention leads to errors.

3. PROBLEM STATEMENT

Industry alert are based on manual intervention. Notification for any circumstances in Industry not provided. Appropriate action for this condition taking. To build the system which can monitor the sensor data and upload it over internet and also capable of taking some crucial decision within industries using the IOT. The IOT consists of objects have less power. They are more dynamic and mobile compare to current state. However, users rely and believe that network will function properly. To maintain robust and trust worthy dynamic ubiquitous networking requires redundancy in several levels and ability to automatically adapt to changed conditions depending on the required quality of service so it should be fault tolerance.

Users are fighting with security and privacy issue of current Internet in large extent. When it will be broaden in to ubiquitous networking, there is even more threat of security and personal privacy. Confidentiality, authenticity and trustworthiness of communication partners need to be maintained. Users may want to give things limited service access not allowing them to communicate in uncontrolled manner.

Initially started with retail and logistic application, the IOT is covering very general applications scenario integrating things to the network. It allows objects to collaborate each other and with person. There are heterogeneous devices, application scenarios, data format, and communication network. Each connected objects should be able to adapt the situation where it is now. When a person with smart phone enters home, it should adapt communication mechanism, addressing and localized environment. When it reaches in office environment it should adapt with new situation where the mechanisms available in home can be different. Adaption in many senses should be maintained thus device adaptation is a necessary concept.

Server computing and IOT are typically used at different levels. IOT which is used at lower levels deals mostly with the end devices, their connections and the protocols they use. The computing process is used at higher levels with the end users and deals with handling and storage of data. In other words, IOT works with low latency and, in the industrial automation, real-time environment. On the other hand, the server is connected to the system through unpredictable latencies. Thus, the following question comes to mind “Is it possible to construct a stable and reliable control system while using server computing as a part of it?” If the server is brought closer to the devices, making a more secure and stable connection, there is no doubt that the automation industry will benefit from it. It would enable a centralized control while minimizing costs. But moving parts of the control system is not simple since it can affect the stability, robustness and reliability of the system.

4. SYSTEM DESIGN

4.1 Block Diagram:

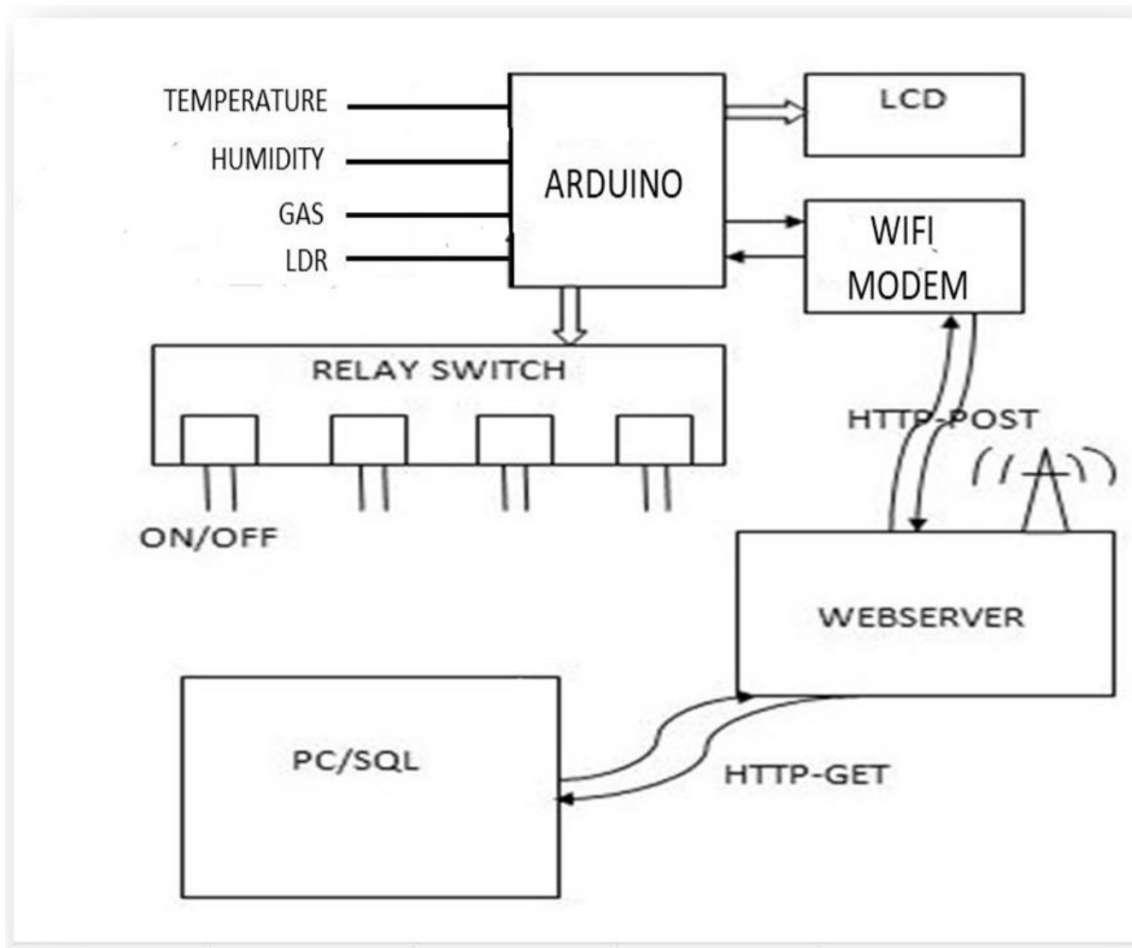


Figure 4.1.1 Block Diagram of Data Acquisition System.

4.2 Components And Specifications:

- **Arduino**

Arduino is open-source hardware and software company, project and user community that designs and manufactures single-board microcontrollers and microcontroller kits for building digital devices and interactive objects that can sense and control objects in the physical and digital world. The boards are equipped with sets of digital and analog input/output (I/O) pins that may be interfaced to various expansion boards or breadboards (shields) and other circuits. The board has 14 Digital pins, 6 Analog pins, and programmable with the Arduino IDE (Integrated Development Environment) via a type B USB cable. It also supports serial communication using TX and Rx pins. It can be powered by a USB cable or by an external 9 volt battery, though it accepts voltages between 7 and 20 volts a 16 MHz quartz crystal, a

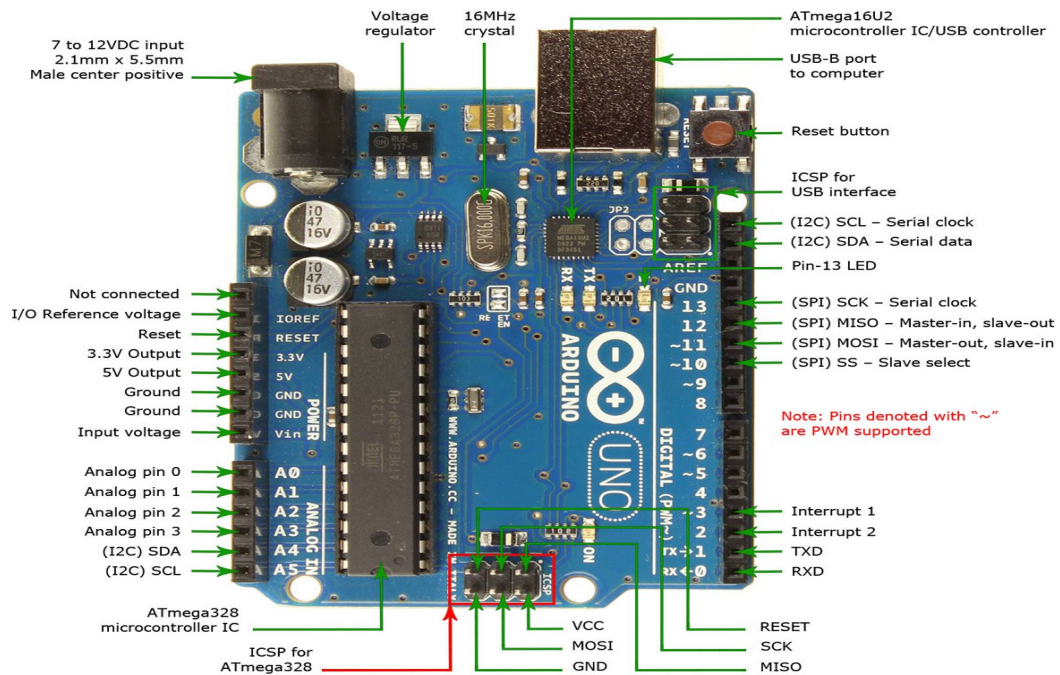


Figure 4.2.1 Pin Diagram of Arduino UNO ATmega328P

USB connection, a power jack, an ICSP header and a reset button. ATmega328P contains everything needed to support the microcontroller. It is connected to computer using a USB cable or given a power supply with AC-to-DC adapter. In case of any severe damage, it can be replaced with a new chip.

Technical specs

- Microcontroller :ATmega328P
- Digital I/O Pins :14 (of which 6 provide PWM output)
- PWM Digital I/O Pins : 6
- Analog Input Pins : 6
- Operating Voltage : 5V
- Input Voltage : 7-12V
- Input Voltage (limit) : 6-20V
- Flash Memory : 32KB (of which 0.5 KB used by boot loader)

- SRAM : 2 KB (ATmega328P)
- EEPROM : 1 KB (ATmega328P)
- DC Current per I/O Pin: 20 mA
- DC Current for 3.3V Pin: 50 mA
- Clock Speed : 16 MHz

- **Temperature Sensor**

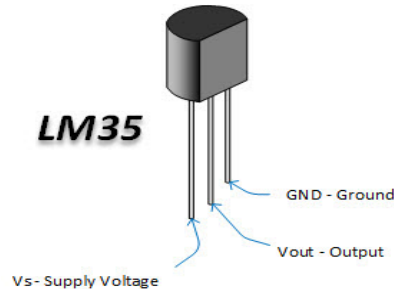


Figure 4.2.2 Temperature Sensor Module.

A temperature sensor is a thermocouple or a resistance temperature detector (RTD) that gathers the temperature from a specific source and alters the collected information into understandable type for an apparatus or an observer. The LM35 series are precision integrated-circuit temperature devices with an output voltage linearly-proportional to the Centigrade temperature. The LM35 device has an advantage over linear temperature sensors calibrated in Kelvin, as the user is not required to subtract a large constant voltage from the output to obtain convenient Centigrade scaling. In general, a temperature sensor is a device which is designed specifically to measure the hotness or coldness of an object. LM35 is a precision IC temperature sensor with its output proportional to the temperature (in °C). With LM35, the temperature can be measured more accurately than with a thermostat.

- **Humidity Sensor**

Stickiness is the nearness of water in air. The measure of water vapor in air can influence human solace and also numerous assembling forms in ventures. The nearness of water vapor likewise impacts different physical, compound, and natural procedures. Mugginess estimation in ventures is basic since it might influence the business cost of the item and the wellbeing and security of the work force.

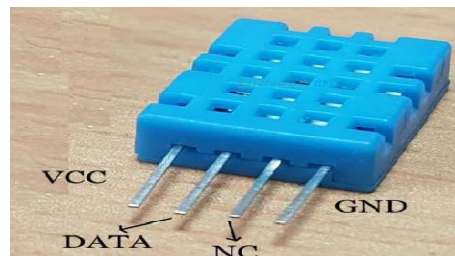


Figure 4.2.3 Humidity Sensor Module

Henceforth, moistness detecting is critical, particularly in the control frameworks for mechanical procedures and human solace. Controlling or observing mugginess is of vital significance in numerous mechanical and household applications. In semiconductor industry,

dampness or dampness levels should be appropriately controlled and observed amid wafer handling. In restorative applications, dampness control is required for respiratory equipment's, sterilizers, hatcheries, pharmaceutical preparing, and organic items. Moistness control is likewise essential in compound gas purging, dryers, stoves, film drying up, paper and material generation, and nourishment handling. In farming, estimation of mugginess is essential for ranch insurance (dew avoidance), soil dampness observing, and so forth. For local applications, mugginess control is required for living condition in structures, cooking control for microwaves, and so on. In every single such application and numerous others, dampness sensors are utilized to give a sign of the dampness levels in the environment. Signal Conditioning Circuits.

- **Gas Sensor**



Figure 4.2.4 Gas Sensor Module.

The MQ6 (LPG Gas Sensor) is a simple-to-use liquefied petroleum gas (LPG) sensor. It can be used in gas leakage detecting equipment in consumer and industry applications, this sensor is suitable for detecting LPG, iso-butane, propane, LNG. Avoid the noise of alcohol, cooking fumes and cigarette smoke. The sensitivity can be adjusted by the potentiometer. Easy to use liquefied petroleum gas (LPG) sensor. High sensitivity to LPG, iso-butane, propane. The MQ-6 can detect gas concentrations anywhere from 200 to 10000ppm. This sensor has a high sensitivity and fast response time. An LPG gas sensor is a one kind of device which is used to sense the presence of a hazardous LPG gas leak in service station, cars, storage tanks and homes. This sensor is attached to an alarm circuit to give an alert to the operators through a buzzer sound in the area where the gas leak is occurring.

- **Light Sensor**



Figure 4.2.5 Light Sensor Module

LDR stands for Light Dependent Resistor or Photo resistor, which is a passive electronic component, basically a resistor which has a resistance that varies depending of the light intensity. A photo resistor is made of a high resistance semiconductor that absorbs photons and based on the quantity and frequency of the absorbed photons the semiconductor material give bound electrons enough energy to jump into the conduction band. There salting free electrons conduct electricity resulting in lowering resistance of the photo resistor. The number

of electrons is dependent of the photons frequency. A Light Sensor generates an output signal indicating the intensity of light by measuring the radiant energy that exists in a very narrow range of frequencies basically called “light”, and which ranges in frequency from “Infra-red” to “Visible” up to “Ultraviolet” light spectrum.

- **WIFI Module**

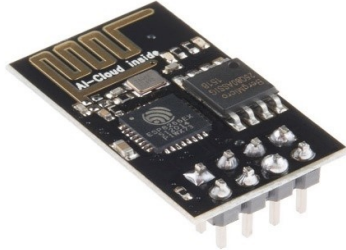


Figure 4.2.6 WIFI Module.

ESP8266EX (simply referred to as ESP8266) is a system-on-chip which integrates a 32-bit ten silica microcontroller; standard digital peripheral interfaces, antenna switches, RF balloon, power amplifier, and low noise receive amplifier, filters and power management modules into a small package. It provides capabilities for 2.4 GHz WI-Fi and pulse width modulation. ESP8266 is Wi-Fi enabled system on chip (SOC) module developed by Espressif system. It is mostly used for development of IOT (Internet of Things) embedded applications. ESP8266 module is low cost standalone wireless transceiver that can be used for end-point IOT developments. To communicate with the ESP8266 module, microcontroller needs to use set of AT commands. Microcontroller communicates with ESP8266-01 module using UART having specified Baud rate.

Applications:

- Internet of Things (IOT)e.g. home automation
- Wireless Remote Control
- Smart plug and lights
- Wireless Sensor Networks
- IP cameras
- Wireless UART

Features:

- Built in Client-Server Mode
- Integrated 10 bit ADC
- 400 Meters Working Distance
- Integrated TCP/IP protocol stack
- On Board 2.4Ghz Ceramic Antenna
- IPX Connector for optional external antenna

- **LCD Display**

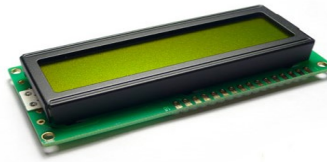


Figure 4.2.7 LCD Display Module.

We come across LCD displays everywhere around us. Computers, calculators, television sets, mobile phones, digital watches use some kind of display to display the time. An LCD is an electronic display module which uses liquid crystal to produce a visible image. The 16×2 LCD display is a very basic module commonly used in DIYs and circuits. The 16×2 translates to a display 16 characters per line in 2 such lines. In this LCD each character is displayed in a 5×7 pixel matrix. LCD (Liquid Crystal Display) screen is an electronic display module and find a wide range of applications. A 16x2 LCD display is very basic module and is very commonly used in various devices and circuits. These modules are preferred over seven segments and other multi segment LEDs. The reasons being: LCDs are economical; easily programmable; have no limitation of displaying special & even custom characters (unlike in seven segments), animations and so on. A 16x2 LCD means it can display 16 characters per line and there are 2 such lines. In this LCD each character is displayed in 5x7 pixel matrix. This LCD has two registers, namely, Command and Data. The command register stores the command instructions given to the LCD. A command is an instruction given to LCD to do a predefined task like initializing it, clearing its screen, setting the cursor position, controlling display etc. The data register stores the data to be displayed on the LCD. The data is the ASCII value of the character to be displayed on the LCD. Click to learn more about internal structure of a LCD.

Pin Description:

Pin No	Function	Name
1	Ground (0V)	Ground
2	Supply voltage; 5V (4.7V – 5.3V)	V _{cc}
3	Contrast adjustment; through a variable resistor	V _{EE}
4	Selects command register when low; and data register when high	Register Select
5	Low to write to the register; High to read from the register	Read/write
6	Sends data to data pins when a high to low pulse is given	Enable
7	8-bit data pins	DB0
8		DB1
9		DB2
10		DB3
11		DB4
12		DB5
13		DB6
14		DB7
15	Backlight V _{CC} (5V)	Led+
16	Backlight Ground (0V)	Led-

Table 4.2.1 LCD pin description

- **RELAYS**



Figure 4.2.8 Relay Module.

Relays are switches that open and close circuits electromechanically or electronically. Relays control one electrical circuit by opening and closing contacts in another circuit. As relay diagrams show, when a relay contact is normally open (NO), there is an open contact when the relay is not energized. When a relay contact is Normally Closed (NC), there is a closed contact when the relay is not energized. In either case, applying electrical current to the contacts will change their state. Relays are generally used to switch smaller currents in a control circuit and do not usually control power consuming devices except for small motors and Solenoids that draw low amps. Nonetheless, relays can "control" larger voltages and amperes by having an amplifying effect because a small voltage applied to a relays coil can result in a large voltage being switched by the contacts. Protective relays can prevent equipment damage by detecting electrical abnormalities, including over current, undercurrent, overloads and reverse currents. In addition, relays are also widely used to switch starting coils, heating elements, pilot lights and audible alarms.

The switch connections of the relays are labelled as COM, NC and NO:

- COM = Common, always connect to this; which is the moving part of the switch.
- NC = Normally Closed, COM is connected to this when the relay coil is off.
- NO = Normally Open, COM is connected to this when the relay coil is on.

Applications of Relay:

- In some types of modems or audio amplifiers relays are used to control high voltage circuit.
- To Control a high-current circuit with a low-current signal, as in the starter solenoid of an automobile.
- Opening and closing circuit breakers detect and isolate faults on transmission and distribution lines.

4.3 Software to Be Used:

- **SQL Software**

Structured Query Language is a domain-specific language used in programming and designed for managing data held in a relational database management system (RDBMS), or for stream processing in a relational data stream management system (RDSMS). It is particularly useful in handling structured data where there are relations between different entities/variables of the data. SQL offers two main advantages over older read/write. SQL implementations are incompatible between vendors and do not necessarily completely follow standards. In particular date and time syntax, string concatenation, null's, and comparison case sensitivity vary from vendor to vendor. Particular exceptions are Postgre SQL and Mimer SQL which

strive for standards compliance, though Postgre SQL does not adhere to the standard in how folding of unquoted names is done. The folding of unquoted names to lower case in Postgre SQL is incompatible with the SQL standard, which says that unquoted names should be folded to upper case. Thus, foo should be equivalent to foo not Foo according to the standard. Popular implementations of SQL commonly omit support for basic features of Standard SQL, such as the date or time data types. The most obvious such examples, and incidentally the most popular commercial and proprietary SQL DBMSs, are Oracle (whose date behaves as date time, and lacks a time type) and MS SQL Server (before the 2008 version). As a result, SQL code can rarely be ported between database systems without modifications.

- **VB.net**

Visual Basic .NET (VB.NET) is a Microsoft object-oriented programming (OOP) language. It evolved from Visual Basic 6 (VB6) to meet an increasing need for easy web-services and web development. VB.Net was designed to take advantage of the .NET framework-based classes and run-time environment. It was re-engineered by Microsoft as part of its .NET product group. Whether Visual Basic .NET should be considered as just another version of Visual Basic or a completely different language is a topic of debate. There are new additions to support new features, such as structured exception handling and short-circuited expressions. Also, two important data-type changes occurred with the move to VB.NET: compared to Visual Basic 6, the integer data type has been doubled in length from 16 bits to 32 bits, and the long data type has been doubled in length from 32 bits to 64 bits. This is true for all versions of VB.NET. A 16-bit integer in all versions of VB.NET is now known as a short. Similarly, the Windows Forms editor is very similar in style and function to the Visual Basic form editor. The things that have changed significantly are the semantics from those of an object-based programming language running on a deterministic, reference-counted engine based on COM to a fully object-oriented language backed by the .NET Framework, which consist of a combination of a common language runtime (a virtual machine using generational garbage collection and a compilation engine) and a far larger class library.

4.4 Working

Data acquisition refers to process of digitizing of analog signals i.e. converting a continuous waveform in time series of digital values which represents signal voltage at that instant.

Once you setup your project, the motion, and vibration of the machine in the industry can be monitored, and for the controlling part, you can turn on the exhaust, and you can turn off or turn on the machines. Although the controlling is done automatically when the sensors detect the change in the environment, you can also control the project using the android app or through the web address. The data is uploaded to the web server, and this data can be seen from the web address or the mobile app you have developed.

- In this project we are going to deal with four sensors like temperature, LDR, gas and humidity which are used to read the physical quantities.
- These four sensors are connected to Arduino ADC pin to convert the digital values in analogy form.
- Program Arduino to sense the data from all the four sensors and send these data through serial communication.
- Arduino reads respective analog values from the sensors and displays the raw data on LCD. then the client read the values serve by the server. After that the controlling action is done automatically.
- On PC we initialize the G.U.I pattern using the desired software. Now wait for incoming packets from server as an output. Thus the incoming packets are passed that display the values on G.U.I. According to the respective values the graph is plotted.
- The data acquired can be stored for that you can create a My SQL database and store the data. Thus the data is then stored in the database as per the conducting information.
- Compare with the limits and generate respective feedback commands. The commands are feedback to the server and according to it the relays make the switches on and off with respect to its sensor.
- This project can send and receive the data within the WI-FI network, in order to access the data and actuate from any part of the world you need to do port forwarding in your network.

5. RESULT AND DISCUSSION

Simulation of the project is done in the Arduino software. The required data conveying the message is transmitted by using server protocol and IOT and the desired output is obtained at receiver.

5.1 Hardware Connection:

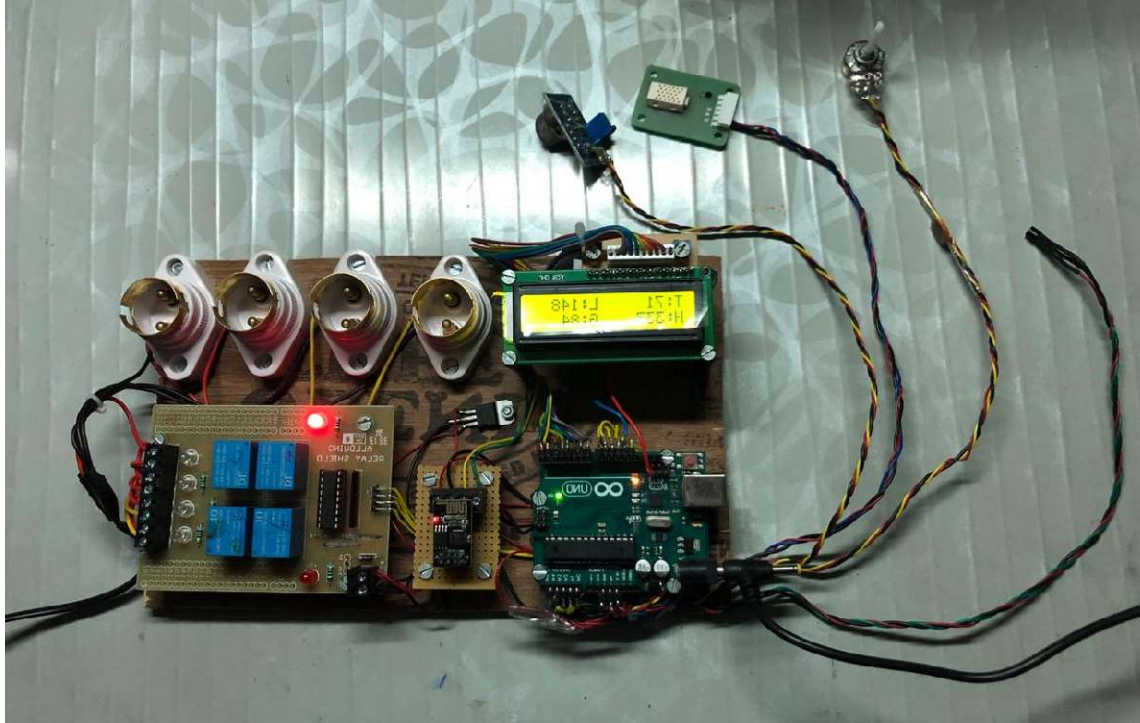


Figure 5.1.1 Experimental Setup

We have incorporated with four sensors namely temperature, humidity, gas and light to read the physical quantities. These four sensors are connected to the Arduino UNO board. Arduino read respective analog values from the sensor. Thus the raw data are displayed on LCD. The data is sending to web server from wifi modem i.e. ESP8266. Client read the values serve by the server and the controlling action is done automatically.

5.2 Output on LCD:

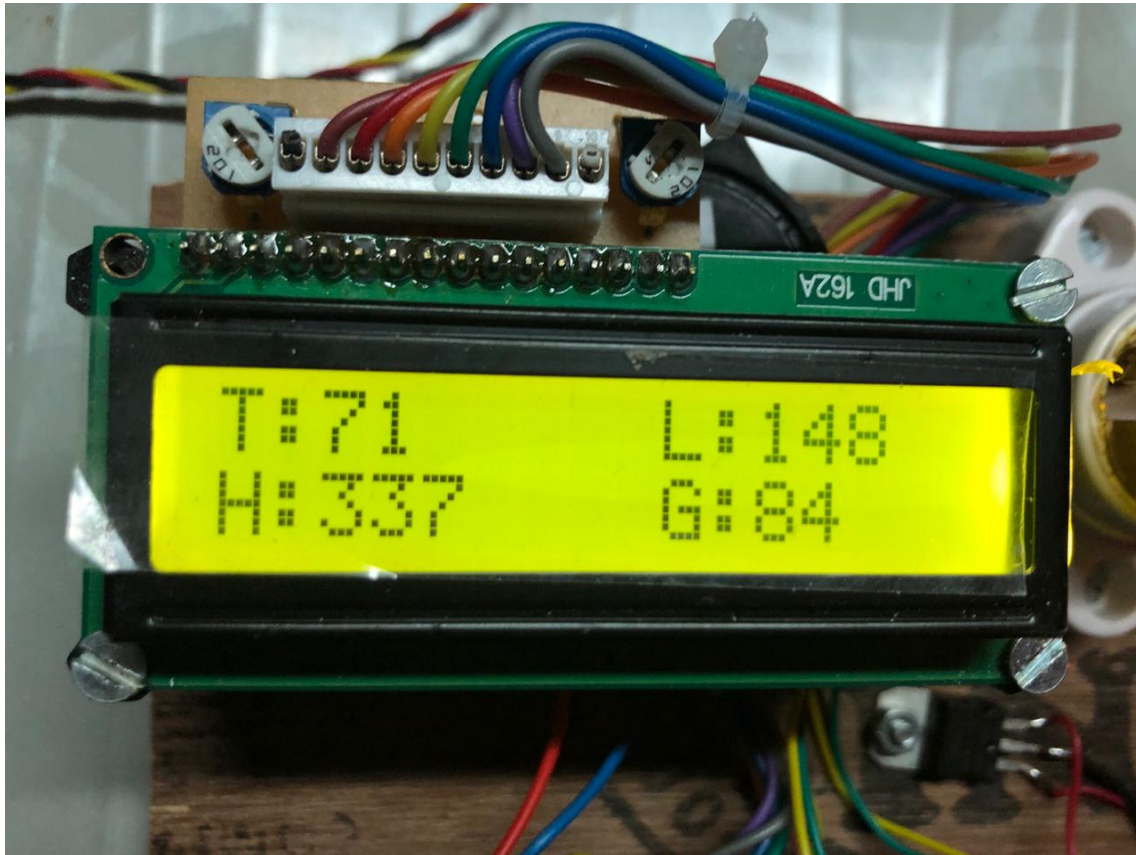


Figure 5.2.1 LCD Output values

The data that is fetch by the Arduino UNO using sensor s are being displayed on the LCD in the form of raw values of the respective parameters.

5.3 VB.Net Output:

The IOT system we have developed is tested in different load conditions. After installing the experimental setup, the user needs to install the software to his/her laptop. After proper installation of the provided software the 16X2 LCD display will show the raw values. After this the address is obtained the user get connected to the setup wifi. As soon as the setup is completed, a page will appear, from which the user could keep a track of all the electronic and electrical devices which are connected with the server.

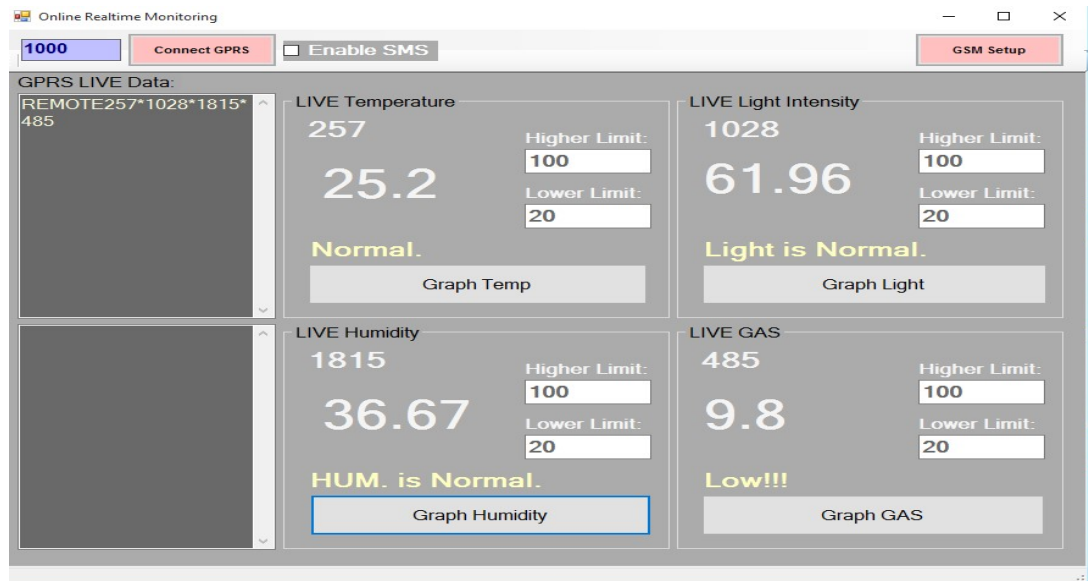


Figure 5.3.1 Output Panel

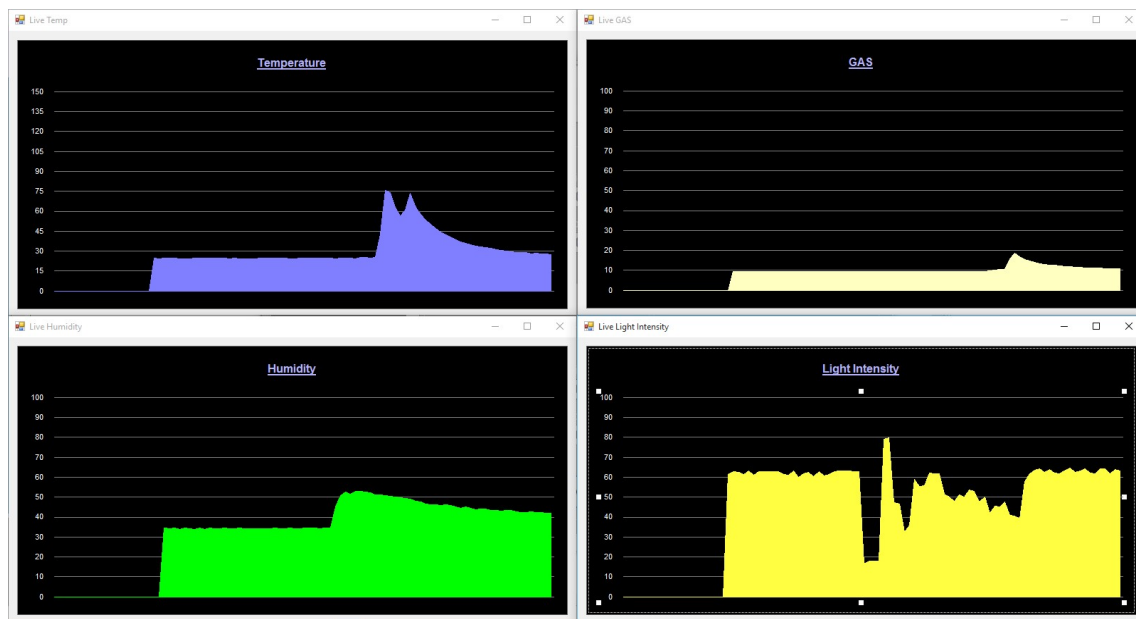


Figure 5.3.2 Output in the Form of GUI

5.4 SQL Output:

The value that has been captured with the help of sensors and the server is recorded in the form of database.

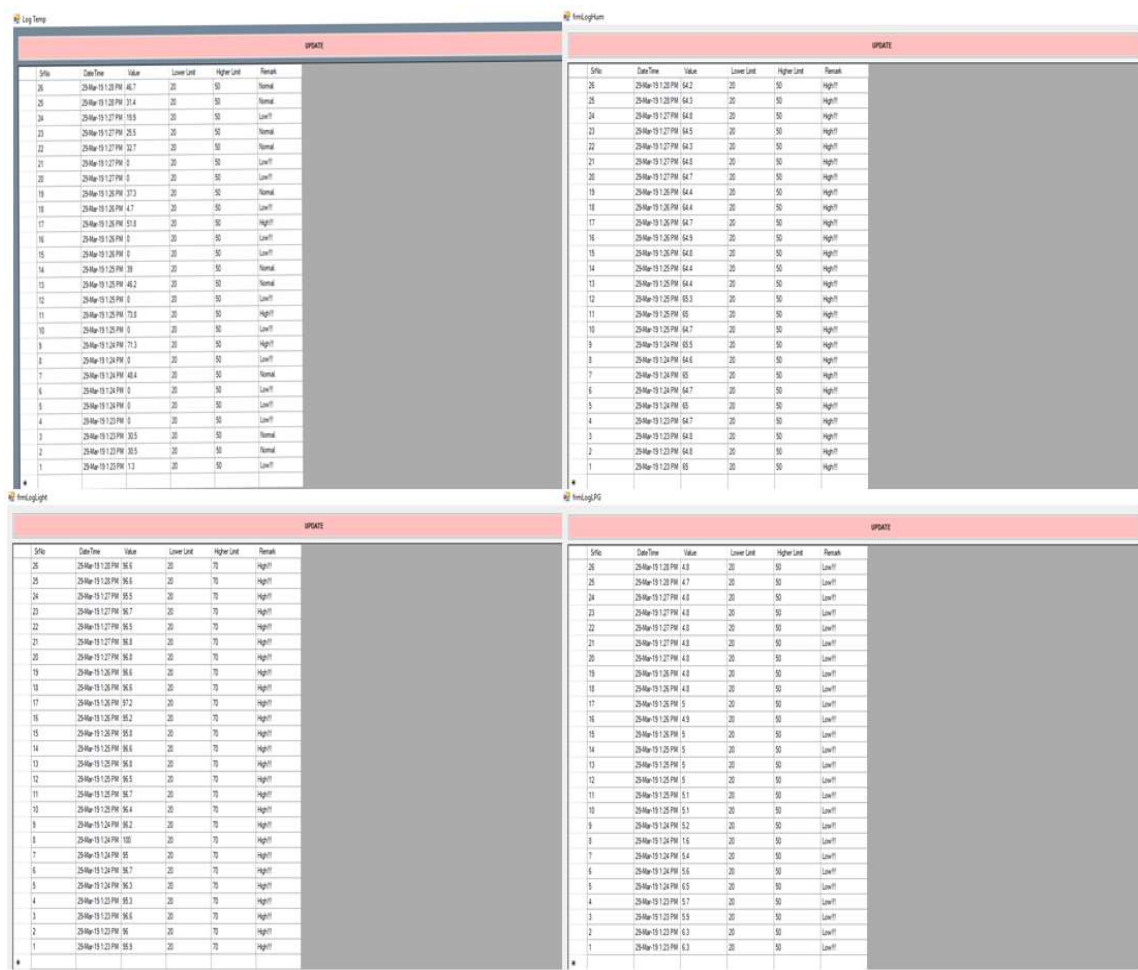


Figure 5.4.1 Database

6. CONCLUSION

6.1 Conclusion:

In this project we focused on different process of operating or controlling electrical and electronic appliances remotely with the help of Arduino. This method of controlling such applications is referred to as automation. The experimental setup which we designed has its focal point on controlling different industrial appliances providing 100% efficiency. Due to advancement in technology, Wi-Fi network is easily available in all places like home, Office Building and Industrial Building so proposed wireless network easily controlled using any Wi-Fi network. The wiring cost is reduced. Since less wiring is required for the switches. This also eliminates power consumption inside the building when the loads were in off conditions.

Our systems, designed with commonly available electronic components, so called hardware; as well as the developed programs, so called software, have worked satisfactorily for recording the physical parameters in real time with graphical display and storing the records to the hard disk of the PC.

Wireless sensors network technology it is one of the most important technologies of 21st century. Smart electronically devices, a large variety of sensors type at affordable price, connected through wireless or internet connections placed in very large numbers, offer the opportunities for monitoring and management applications. All of these networks of micro sensors provide new solutions to many vital areas such as health or industry, offering new options in monitoring actions or other industrial, automotive applications. Sensors can be installed on the ground, in the air, underwater, on the body of persons, in auto vehicles or inside of buildings. All these sensors connected together on a network can be used to process a lot of events, to monitor vehicles, persons or to identify biological or chemical agents. Each intelligent sensor from the local network is based on an embedded system which is capable to process data from one or more sensors implemented with less processing capabilities.

6.2 Future Scope:

The present work is focused around the design and development of cost effective PC based data acquisition system (DAS), both hardware's and software's. To make the designed system more flexible with better performance, the following modifications can be made

- (i) Sensor : selection of high quality sensors, integrated or smart sensors to get faster response speed with higher resolution and better performance,
- (ii) Hardware and software: hardware and software modifications to enable long distance acquisition, etc.
- (iii) Miniaturization: Miniaturization of the designed system by using surface mount devices (SMD components).
- (iv) Adding speech recognition to the system.
- (v) The project can also be monitored from the mobile through an android app, you can use MIT app inventor to create an app.

Then, the scope and applicability of the designed system will be increased. It can be used for the applications in various fields in physics, chemistry, life sciences, engineering, medical, geological applications etc.

7. REFERENCE

Journal papers

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- [2] Stephen Nischay , Latha, “Design of DACS (Data Acquisition and Control System) using Linux” International Journal of Emerging Technology and Advanced Engineering , Volume 2, Issue 11, November 2012, pg. 344 to 346
- [3] Saraswat Teli , Mani , “ Smart Real Time Embedded Arduino Based Data Acquisition System” International Journal of Research in Engineering and Technology, Volume: 04 Issue: 08 , August-2015, pg. 258 to 262

Websites

- [4] https://en.wikipedia.org/wiki/Internet_of_things
- [5] https://en.wikipedia.org/wiki/Data_acquisition
- [6] https://en.wikipedia.org/wiki/List_of_temperature_sensors
- [7] <https://en.wikipedia.org/wiki/ESP8266>

APPENDIX

Arduino Code:

```
#include <LiquidCrystal.h>
LiquidCrystal lcd(2, 3, 4, 5, 6, 7);

//String MySSID = "LogicInside_TPLINK";
//String MyPass = "Earth0509";

String MySSID = "iPhone";
String MyPass = "12345678";

int RL1=8;
int RL2=9;
int RL3=10;
int RL4=11;
int BUZZ=12;
//Functions-----
void Beep(int No, int Delay);

//Variables-----
int i=0;
int RxLEN = 0;
String TxPacket,RxPacket;
unsigned int Temp,Light,Hum,Gas;
//-----
#####
#####
void setup()
{
  //Peripherals-----
  pinMode(RL1, OUTPUT);digitalWrite(RL1, LOW);
  pinMode(RL2, OUTPUT);digitalWrite(RL2, LOW);
  pinMode(RL3, OUTPUT);digitalWrite(RL3, LOW);
  pinMode(RL4, OUTPUT);digitalWrite(RL4, LOW);
  pinMode(BUZZ, OUTPUT);Beep(3,100);
  //LCD-----
  lcd.begin(16, 2);
  lcd.setCursor(0,0);lcd.print("Welcome to.....");
  lcd.setCursor(0,1);lcd.print("  APSIT  ");
  delay(2000);
  lcd.setCursor(0,0);lcd.print("IOT Based Data ");
  lcd.setCursor(0,1);lcd.print("Acquisition Sys.");
  delay(2000);
  lcd.setCursor(0,0);lcd.print("Project Done By.");
  lcd.setCursor(0,1);lcd.print("Musharraf  ");
  delay(2000);
  lcd.setCursor(0,0);lcd.print("Vishakha  ");
  lcd.setCursor(0,1);lcd.print("Vrutika  ");
  delay(2000);
  lcd.setCursor(0,0);lcd.print("Karan  ");
```

```

lcd.setCursor(0,1);lcd.print("*****");
delay(2000);
lcd.setCursor(0,0);lcd.print("Under Guidance..");
lcd.setCursor(0,1);lcd.print("Prof. Mahesh P. ");
delay(2000);

Serial.begin(115200);
espInitIOT(MySSID,MyPass);

lcd.setCursor(0,0);lcd.print("System Started..");
lcd.setCursor(0,1);lcd.print("*****");
delay(2000);

}

void loop()
{
  //Monitor Values
  Temp = analogRead(A0);
  Light = analogRead(A1);
  Hum = analogRead(A2);
  Gas = analogRead(A3);
  //display on LCD
  lcd.clear();

  lcd.setCursor(0,0);lcd.print("T:");lcd.print(Temp);lcd.setCursor(9,0);lcd.print("L:");lcd.print(
  Light);

  lcd.setCursor(0,1);lcd.print("H:");lcd.print(Hum);lcd.setCursor(9,1);lcd.print("G:");lcd.print(
  Gas);

  //send to IOT
  TxPacket = "REMOTE" + String(Temp) + "*" + String(Light) + "*" + String(Hum) + "*" +
  String(Gas);
  espHTTP_POST_STR("LIG_90919HW.txt",TxPacket);
  Beep(1,100);
  delay(2000);

  //get response from IOT
  RxPacket = espHTTP_GET_STR("LIG_90919PC.txt");
  if(RxPacket[0]=='0')digitalWrite(RL1, LOW);
  if(RxPacket[0]=='1')digitalWrite(RL1, HIGH);
  if(RxPacket[1]=='0')digitalWrite(RL2, LOW);
  if(RxPacket[1]=='1')digitalWrite(RL2, HIGH);
  if(RxPacket[2]=='0')digitalWrite(RL3, LOW);
  if(RxPacket[2]=='1')digitalWrite(RL3, HIGH);
  if(RxPacket[3]=='0')digitalWrite(RL4, LOW);
  if(RxPacket[3]=='1')digitalWrite(RL4, HIGH);

  Beep(2,100);
  delay(2000);
}

```


VB.Net Code:

Option Explicit On

Imports System.Net

Public Class frmMain

 'database

 Dim dbCon As New ADODB.Connection

 Dim dbTemp As New ADODB.Recordset

 Dim dbLight As New ADODB.Recordset

 Dim dbHum As New ADODB.Recordset

 Dim dbLPG As New ADODB.Recordset

 Private Sub frmMain_Load(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles MyBase.Load

 dbCon.Open("Driver={SQL Server}; Server=.\SQLEXPRESS;Database=SCADA",)

 End Sub

 Private Sub btnGPRS_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles btnGPRS.Click

 tmrLIVERx.Interval = Val(txtDelay.Text)

 tmrLIVERx.Enabled = True

 End Sub

 Private Sub tmrLIVERx_Tick(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles tmrLIVERx.Tick

 On Error GoTo X

 Dim LIVE As New WebClient

 tmrLIVERx.Interval = Val(txtDelay.Text)

 Dim strLIVE As String

 strLIVE

 LIVE.DownloadString("http://IOT.logicinside.net/LIG_90919HW.txt").ToCharArray

 txtLIVE.Text = strLIVE

 If strLIVE.StartsWith("NOTHING") Then

 Exit Sub

 End If

 If strLIVE.StartsWith("REMOTE") Then

 Dim strDATA As String

 strDATA = strLIVE.Substring(6)

 'data received-----

 Dim strParsed As String()

 strParsed = strDATA.Split("*")

 Dim RByte As Integer

 Dim temp As Single


```

Dim R As Integer

Dim strRES As String
strRES = ""

'Temperature////////////////////////////////////////
Dim Temperature As Single
'display value-----
RByte = Val(strParsed(0))
lblTempRaw.Text = RByte
Temperature = Format(RByte / 2.3571, "00.0")
lblTemp.Text = Temperature
'graph plotting-----
'shift all data
For R = 1 To frmTemp.Chart.RowCount - 1
    frmTemp.Chart.Row = R + 1
    temp = frmTemp.Chart.Data
    frmTemp.Chart.Row = R
    frmTemp.Chart.Data = temp
Next R
'insert data in last row
frmTemp.Chart.Row = frmTemp.Chart.RowCount
frmTemp.Chart.Data = Temperature
'Limit chk-----
If Temperature < Val(txtTempLL.Text) Then
    lblTempMsg.Text = "Low!!!"
    strRES = strRES & "1"
End If
If Temperature > Val(txtTempHL.Text) Then
    lblTempMsg.Text = "High!!!"
    strRES = strRES & "0"
End If
If Temperature >= Val(txtTempLL.Text) And Temperature <= Val(txtTempHL.Text)
Then
    lblTempMsg.Text = "Normal."
    strRES = strRES & "X"
End If
'database-----
If chkDB.Checked = True Then
    Dim SrNo As Integer
    dbTemp.Open("SELECT * FROM Temp", dbCon,
ADODB.CursorTypeEnum.adOpenDynamic, ADODB.LockTypeEnum.adLockOptimistic)
    If dbTemp.BOF Then
        SrNo = 1
    Else
        dbTemp.MoveLast()
    End If
End If

```

```

        SrNo = dbTemp.Fields("SrNo").Value + 1
    End If
    dbTemp.AddNew()
    dbTemp.Fields("SrNo").Value = SrNo
    dbTemp.Fields("DateTime").Value = DateTime.Now
    dbTemp.Fields("Value").Value = lblTemp.Text
    dbTemp.Fields("Lower Limit").Value = Val(txtTempLL.Text)
    dbTemp.Fields("Higher Limit").Value = Val(txtTempHL.Text)
    dbTemp.Fields("Remark").Value = lblTempMsg.Text
    dbTemp.Update()
    dbTemp.Close()
End If

```

```

'Light Intensity////////////////////////////////////////
Dim Light As Single
'display value-----
RByte = Val(strParsed(1))
lblLightRaw.Text = RByte
Light = Format((1025 - RByte) / 10.25, "00.0")
lblLight.Text = Light
'graph plotting-----
'shift all data
For R = 1 To frmLight.Chart.RowCount - 1
    frmLight.Chart.Row = R + 1
    temp = frmLight.Chart.Data
    frmLight.Chart.Row = R
    frmLight.Chart.Data = temp
Next R
'insert data in last row
frmLight.Chart.Row = frmLight.Chart.RowCount
frmLight.Chart.Data = Light
'Limit chk-----
If Light < Val(txtLightLL.Text) Then
    lblLightMsg.Text = "Low!!!"
    strRES = strRES & "1"
End If
If Light > Val(txtLightHL.Text) Then
    lblLightMsg.Text = "High!!!"
    strRES = strRES & "0"
End If
If Light >= Val(txtLightLL.Text) And Light <= Val(txtLightHL.Text) Then
    lblLightMsg.Text = "Normal."
    strRES = strRES & "X"
End If

```

```

'database-----
If chkDB.Checked = True Then
    Dim SrNo As Integer
    dbLight.Open("SELECT * FROM Light", dbCon,
ADODB.CursorTypeEnum.adOpenDynamic, ADODB.LockTypeEnum.adLockOptimistic)
    If dbLight.BOF Then
        SrNo = 1
    Else
        dbLight.MoveLast()
        SrNo = dbLight.Fields("SrNo").Value + 1
    End If
    dbLight.AddNew()
    dbLight.Fields("SrNo").Value = SrNo
    dbLight.Fields("DateTime").Value = DateTime.Now
    dbLight.Fields("Value").Value = lblLight.Text
    dbLight.Fields("Lower Limit").Value = Val(txtLightLL.Text)
    dbLight.Fields("Higher Limit").Value = Val(txtLightHL.Text)
    dbLight.Fields("Remark").Value = lblLightMsg.Text
    dbLight.Update()
    dbLight.Close()
End If

'Humidity////////////////////////////////////////
Dim Hum As Single
'display value-----
RByte = Val(strParsed(2))
lblHumRaw.Text = RByte
Hum = Format((1025 - RByte) / 10.25, "00.0")
lblHum.Text = Hum
'graph plotting-----
'shift all data
For R = 1 To frmHum.Chart.RowCount - 1
    frmHum.Chart.Row = R + 1
    temp = frmHum.Chart.Data
    frmHum.Chart.Row = R
    frmHum.Chart.Data = temp
Next R
'insert data in last row
frmHum.Chart.Row = frmHum.Chart.RowCount
frmHum.Chart.Data = Hum
'Limit chk-----
If Hum < Val(txtHumLL.Text) Then
    lblHumMsg.Text = "Low!!!"
    strRES = strRES & "1"
End If
If Hum > Val(txtHumHL.Text) Then

```

```

        lblHumMsg.Text = "High!!!"
        strRES = strRES & "0"
    End If
    If Hum >= Val(txtHumLL.Text) And Hum <= Val(txtHumHL.Text) Then
        lblHumMsg.Text = "Normal."
        strRES = strRES & "X"
    End If
    'database-----
    If chkDB.Checked = True Then
        Dim SrNo As Integer
        dbHum.Open("SELECT * FROM Humidity", dbCon,
ADODB.CursorTypeEnum.adOpenDynamic, ADODB.LockTypeEnum.adLockOptimistic)
        If dbHum.BOF Then
            SrNo = 1
        Else
            dbHum.MoveLast()
            SrNo = dbHum.Fields("SrNo").Value + 1
        End If
        dbHum.AddNew()
        dbHum.Fields("SrNo").Value = SrNo
        dbHum.Fields("DateTime").Value = DateTime.Now
        dbHum.Fields("Value").Value = lblHum.Text
        dbHum.Fields("Lower Limit").Value = Val(txtHumLL.Text)
        dbHum.Fields("Higher Limit").Value = Val(txtHumHL.Text)
        dbHum.Fields("Remark").Value = lblHumMsg.Text
        dbHum.Update()
        dbHum.Close()
    End If

    'GAS////////////////////////////////////
    Dim GAS As Single
    'display value-----
    RByte = Val(strParsed(3))
    lblGASRaw.Text = RByte
    GAS = Format(RByte / 10.25, "00.0")
    lblGAS.Text = GAS
    'graph plotting-----
    'shift all data
    For R = 1 To frmGAS.Chart.RowCount - 1
        frmGAS.Chart.Row = R + 1
        temp = frmGAS.Chart.Data
        frmGAS.Chart.Row = R
        frmGAS.Chart.Data = temp
    Next R
    'insert data in last row
    frmGAS.Chart.Row = frmGAS.Chart.RowCount

```

```

frmGAS.Chart.Data = GAS
'Limit chk-----
If GAS < Val(txtGASLL.Text) Then
    lblGASMsg.Text = "Low!!!"
    strRES = strRES & "1"
End If
If GAS > Val(txtGASHL.Text) Then
    lblGASMsg.Text = "High!!!"
    strRES = strRES & "0"
End If
If GAS >= Val(txtGASLL.Text) And GAS <= Val(txtGASHL.Text) Then
    lblGASMsg.Text = "Normal."
    strRES = strRES & "X"
End If
'database-----
If chkDB.Checked = True Then
    Dim SrNo As Integer
    dbLPG.Open("SELECT * FROM LPG", dbCon,
ADODB.CursorTypeEnum.adOpenDynamic, ADODB.LockTypeEnum.adLockOptimistic)
    If dbLPG.BOF Then
        SrNo = 1
    Else
        dbLPG.MoveLast()
        SrNo = dbLPG.Fields("SrNo").Value + 1
    End If
    dbLPG.AddNew()
    dbLPG.Fields("SrNo").Value = SrNo
    dbLPG.Fields("DateTime").Value = DateTime.Now
    dbLPG.Fields("Value").Value = lblGAS.Text
    dbLPG.Fields("Lower Limit").Value = Val(txtGASLL.Text)
    dbLPG.Fields("Higher Limit").Value = Val(txtGASHL.Text)
    dbLPG.Fields("Remark").Value = lblGASMsg.Text
    dbLPG.Update()
    dbLPG.Close()
End If

'Mark that data has been fetched
LIVE.UploadString("http://IOT.logicinside.net", "LIG_90919HW.txt" &
"NOTHING")

'send the swtching command to hw
LIVE.UploadString("http://IOT.logicinside.net", "LIG_90919PC.txt" & "~" &
strRES)

End If

```

```

Exit Sub
X:
    txtSysMsg.Text = Err.Description
End Sub
Private Sub btnClearLIVE_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs)
    txtLIVE.Text = ""
End Sub
Private Sub btnGraphTemp_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles btnGraphTemp.Click
    frmTemp.Show()
End Sub

Private Sub btnGraphLight_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles btnGraphLight.Click
    frmLight.Show()
End Sub

Private Sub btnGraphHum_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles btnGraphHum.Click
    frmHum.Show()
End Sub

Private Sub btnGraphLPG_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles btnGraphGAS.Click
    frmGAS.Show()
End Sub

Private Sub btnLogTemp_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles btnLogTemp.Click
    frmLogTemp.Show()
End Sub

Private Sub btnLogLight_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles btnLogLight.Click
    frmLogLight.Show()
End Sub

Private Sub btnLogHum_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles btnLogHum.Click
    frmLogHum.Show()
End Sub
Private Sub btnLogLPG_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles btnLogGAS.Click
    frmLogGAS.Show()

```

End Sub
End Class